

# Multi-Agent Reinforcement Learning for Visual Comfort Enhancement of Casual Stereoscopic Photography

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## ABSTRACT

The goal of casual stereoscopic photography is to allow ordinary users to create a stereoscopic photo using two photos taken casually by a monocular camera. In this poster, we propose a multi-agent reinforcement learning framework for visual comfort enhancement of casual stereoscopic photography. Each agent calculates a homography matrix based on the positions of four corners before and after the offsets. Furthermore, we propose a hierarchical stereo transformer based on window attention, which enhances and fuses multiscale correlations between left and right views. Experimental results show that our proposed method achieves superior performance to the state-of-the-art methods.

**Index Terms:** Human-centered computing—3D authoring—Image and video acquisition—Computer vision;

## 1 INTRODUCTION

Casual stereoscopic photography allows common users to create a stereoscopic photo using two photos taken casually with a hand-held monocular camera. Specifically, one can first take a photo, move the camera horizontally, and take a second photo. However, during taking photos for casual stereoscopic photography, the captured image pair can hardly satisfy stereoscopic geometric constraints. Furthermore, it is also difficult for ordinary users to adjust the disparity and gain a high-quality stereoscopic image with satisfactory image composition and visual comfort [2].

Traditional approaches regarding visual comfort enhancement (VCE) of stereo images have focused on two types of enhancement: stereo correction and disparity adjustment. Stereo correction methods, aiming at satisfying stereo geometric constraints, can be divided into calibrated methods [4] and uncalibrated methods [1, 3]. Inspired by the sequential decision-making process of human experts, previous works [2, 9] propose reinforcement learning-based visual comfort enhancement methods for casual stereoscopic photography. Specifically, they design model actions that can simulate camera motions and use a reinforcement learning framework to choose actions to adjust left and right views of the stereoscopic image and gradually improve the visual comfort at each step.

In this poster, we present a multi-agent reinforcement learning framework to handle the coordination of left and right views in the process of visual comfort enhancement. To this end, we define actions through a unified view transformation method, which directly calculates a homography matrix from the offsets of four corners of each view. Compared with the specially designed actions of previous works, our method has better generalizability and flexibility. Furthermore, we present a hierarchical window-based stereo transformer to

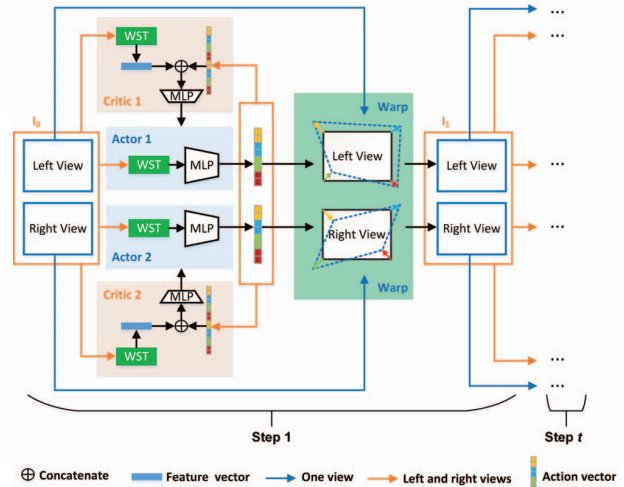


Figure 1: Illustration of our multi-agent reinforcement-learning framework for visual comfort enhancement.

enhance and integrate low-level and high-level features from left and right views. Experimental results on synthetic datasets demonstrate that our method outperforms the existing VCE methods. And a user study on real stereoscopic images shows that participants generally prefer our results.

## 2 OUR PROPOSED METHOD

### 2.1 Multi-agent reinforcement learning for VCE

As shown in Fig. 1, each actor and critic contains a Window-based Stereo Transformer (WST) for feature fusion and a multi-layer perceptron (MLP). At each step, Actor 1 and Actor 2 jointly decide actions based on the input left and right views, and the action of each agent is an eight-dimensional action vector representing offsets of four corners. The homography matrix corresponding to the transformation of each view can be calculated from the four corners' positions and offsets. Then, two homography matrices are applied to the input stereo images to obtain the transformed left and right views which serve as input in the next iteration. Two critics are used to evaluate two agents based on the input stereo images and action vectors. This process will be iteratively performed for  $t$  times and produces final results.

### 2.2 Hierarchical window-based stereo transformer

Inspired by Swin Transformer [8], we propose a hierarchical window-based stereo transformer to handle the dense correspondence at pixel level within windows of left and right views. Following Swin Transformer, a window-based stereo transformer is a three-stage hierarchical structure. At stage one, left and right views first go through patch partition and linear embedding to get patch embedding respectively. The patch embedding is then fed to three successive stereo attention blocks to obtain the output of the first stage. Both stage two and stage three consist of three successive

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Table 1: Geometric transformation evaluation. The best performance values are highlighted in bold.

| Dataset            | Syn-A        |               |              | Syn-B        |               |              |
|--------------------|--------------|---------------|--------------|--------------|---------------|--------------|
| Metric             | VCA          | PSNR          | SSIM         | VCA          | PSNR          | SSIM         |
| Calib. Rect. [4]   | 3.194        | 14.231        | 0.378        | 3.196        | 12.986        | 0.181        |
| Uncalib. Rect. [1] | 3.514        | 17.741        | 0.496        | 3.506        | 14.864        | 0.260        |
| Q-E Rect. [3]      | 3.471        | 16.909        | 0.503        | 3.553        | 15.999        | 0.303        |
| Geo. Net [5]       | 3.706        | 24.882        | 0.715        | 3.738        | 19.358        | 0.420        |
| Recur.-VCE [9]     | 3.759        | 24.296        | 0.711        | 3.810        | 19.610        | 0.447        |
| Conti.-VCE [2]     | 3.868        | 28.210        | 0.804        | 3.814        | 21.061        | 0.496        |
| Ours               | <b>3.876</b> | <b>31.120</b> | <b>0.862</b> | <b>3.846</b> | <b>21.244</b> | <b>0.515</b> |

stereo attention blocks. Between stages, the number of patches is reduced by patch merging, where the features of each group of  $2 \times 2$  neighboring patches are concatenated in the same way as patch merging in Swin Transformer.

A stereo attention block consists of multi-head self-attention (MSA) modules and multi-head cross-attention (MCA) modules. Firstly, we utilize MSA modules to enhance features of left and right views, respectively, by calculating the attention among features within a window. Then, MCA modules are used to fuse the enhanced features of left and right views by computing the attention between the corresponding window of left and right views.

### 3 EXPERIMENTS

#### 3.1 Datasets and preprocessing

To train and evaluate our proposed method, we synthesize low-quality stereoscopic images from three high-quality public stereo datasets (i.e., IVY stereoscopic video dataset [6], IVY stereoscopic image dataset [7] and IEEE-SA stereo image dataset [10]), then use the synthesized images as training data and the original high-quality stereoscopic images as reference images. Specifically, we apply geometric transformations on IVY stereoscopic video dataset and IVY stereoscopic image dataset to synthesize low-quality stereoscopic image datasets, denoted as *Syn-A* and *Syn-B*, respectively. Then, we use the IEEE-SA stereo image dataset to generate a dataset that does not satisfy geometric constraints and is not at an appropriate disparity, denoted as *Syn-C-Mix*.

#### 3.2 Geometric transformation

As shown in Table 1, both PSNR and SSIM metrics of our method on the *Syn-A* dataset outperform the compared methods, indicating that our methods are more consistent with ground truths. Meanwhile, the VCA metric of our method on the *Syn-A* dataset is better than the compared methods, indicating that our results have better visual comfort compared to the compared methods. In addition, since the *Syn-B* dataset is only used for testing and our method outperforms the compared methods in VCA, PSNR, and SSIM metrics, indicating that our method has good generalization capability.

#### 3.3 Casual stereoscopic photography

To evaluate the VCE capability of our proposed method for casual stereoscopic photography, we experiment on the dataset *Syn-C-Mix*, whose images violate geometric constraints and exceed comfort disparity ranges. In this subsection, we compare our method with two reinforcement learning-based visual comfort enhancement methods [2, 9]. We demonstrate the quantitative comparison results of casual stereoscopic photography on *Syn-C-Mix* in Table 2.

#### 3.4 User study on real images

We also conduct a user study to evaluate our method on real-world images. To this end, we compare our method with Recurrent-VCE [9] and Continuous-VCE [2] on 20 real-world stereoscopic images captured by monocular mobile cameras. In practice, we recruit 10 participants (5 males and 5 females) to view 20 comparison pairs between our results and those of [9], and 20 comparison pairs

Table 2: Casual stereoscopic photography evaluation.

| Metric             | VCA          | PSNR          | SSIM         |
|--------------------|--------------|---------------|--------------|
| Recurrent-VCE [9]  | 3.487        | 16.956        | 0.333        |
| Continuous-VCE [2] | 3.792        | 20.574        | 0.477        |
| Ours               | <b>3.799</b> | <b>20.756</b> | <b>0.505</b> |

between our results and those of [2]. Following [9], for each comparison pair, the participants are asked to perform the following tasks: (1) select the preferred image that is more visually comfortable, and (2) select the preferred one with better general quality, including visual comfort, sense of 3D presence, etc.

For each participant, we calculate the times of selecting our results rather than those of comparison methods in the 20 comparisons. On average, the percentage values that users prefer our results over the results of [9] with respect to visual comfort and general quality are 87.5% and 82.5%, respectively. And the percentage values that users prefer our results over the results of [2] with respect to visual comfort and general quality are 86.0% and 81.5%, respectively.

### 4 CONCLUSION AND LIMITATIONS

In this poster, we propose a multi-agent reinforcement learning framework for visual comfort enhancement of casual stereoscopic photography. Specifically, each agent calculates a transformation matrix based on the predicted offsets of its four corners and then applies the transformation matrix to the current view to get the view for the next iteration. In addition, we propose a hierarchical stereo transformer based on window attention, which enhances and fuses multiscale correlations between left and right views. Finally, we conduct comprehensive experiments to validate the effectiveness of our proposed method.

Our framework still has several limitations. Directly predicting consecutive offset values of four points leads to an oversized solution space for the model and also makes training relatively difficult. In addition, training our model requires a large amount of data, and there is still a gap between real data and our synthetic data.

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